

Safety Data Sheet

Product name: DL-Methionine 99% Feed Grade

Product code: SLDL01

Current version: 3.0, issued: Jul. 30th, 2022 **Replaced version:** 2.0, issued: Jun. 8th, 2022

Section 1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier:

Product name: **DL-Methionine 99% Feed Grade**
Product code: SLDL01
Brand: NHU
CAS No.: 59-51-8
EC No.: 200-432-1

1.2 Relevant identified uses of the substance or mixture and uses advised against:

Relevant identified uses of the substance or mixture:

Feed additives.

1.3 Details of the supplier of the safety data sheet:

Company

Shandong NHU Amino acid Co., Ltd.

Address

No.02999 Zhujiang West Third Road, Yangzi Street.
Binhai District, Weifang City, Shandong Province, China.

Tel: 0575-86124428

Fax: 0575-86128321

E-Mail: sales@cnhu.com

Website: <http://www.cnhu.com/>

1.4 Emergency telephone number:

Please contact the regional representative in your country.

Section 2. Hazard identification

2.1 Classification of the substance or mixture:

Classification information:

This product does not meet the classification and labelling criteria given in the Regulation (EC) No 1272/2008 (CLP).

2.2 Label elements:

Not relevant.

2.3 Other hazards:

No data available.

Section 3. Composition and information on ingredients

3.1 Substances:

Chemical characterization

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Substance name DL-methionine

Degree of purity $\geq 99\%$

Formula $C_5H_{11}O_2NS$

Molecular weight 149.21

Identification numbers

CAS no. 59-51-8

EC no. 200-432-1

3.2 Mixtures:

Not applicable. The product is not a mixture.

Section 4. First aid measures

4.1 Description of first aid measures:

General information:

Remove contaminated clothing and shoes and launder thoroughly before reusing. No special measures necessary.

After inhalation:

Ensure supply of fresh air. Irregular breathing/no breathing: artificial respiration.

After skin contact:

In case of contact with skin wash off immediately with soap and water.

After eye contact:

Remove contact lenses, irrigate copiously with clean, fresh water for at least 15 minutes, holding the eyelids apart and seek medical advice. In case of irritation consult an ophthalmologist.

After ingestion:

Induce vomiting if patient is conscious, seek medical advice. Rinse mouth thoroughly with water.

4.2 Most important symptoms and effects, both acute and delayed:

No data available.

4.3 Indication of any immediate medical attention and special treatment needed:

No data available.

Section 5. Fire-fighting measures

5.1 Extinguishing media:

Suitable extinguishing media :

Alcohol-resistant foam; Extinguishing powder; Water spray jet; Carbon dioxide.

Unsuitable extinguishing media:

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High power water jet.

5.2 Special hazards arising from the substance or mixture:

In the event of fire, the following can be released: Carbon dioxide (CO₂); Carbon monoxide (CO); Nitrogen oxides (NO_x); Sulphur oxides.

5.3 Advice for firefighters:

Fire-fighting operations, rescue and clearing work under effect of combustion and smoulder gases just may be done with breathing apparatus. Wear protective clothing.

Section 6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures:

For non-emergency personnel

Ensure adequate ventilation. Avoid dust formation. Remove persons to safety.

For emergency responders

No data available. Personal protective equipment (PPE) - see Section 8.

6.2 Environmental precautions:

Do not discharge into the drains/surface waters/groundwater.

6.3 Methods and material for containment and cleaning up:

Avoid raising dust. Take up mechanically. When picked up, treat material as prescribed under heading "Disposal considerations".

6.4 Reference to other sections:

Information regarding waste disposal, see chapter 13. Information regarding personal protective measures, see chapter 8.

Information regarding safe handling, see chapter 7.

Section 7. Handling and storage

7.1 Precautions for safe handling:

Advice on safe handling

Provide good ventilation at the work area (local exhaust ventilation, if necessary).

General protective and hygiene measures

Keep away from food, drink and animal feeding stuffs. Avoid contact with eyes and skin. Do not inhale dust. Wash hands before breaks and after work. Remove soiled or soaked clothing immediately.

Advice on protection against fire and explosion

Keep away from sources of ignition - refrain from smoking. Take precautionary measures against static charges. Dust can form an explosive mixture with air.

7.2 Conditions for safe storage, including any incompatibilities:

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Technical measures and storage conditions

Protect from light. Keep container tightly closed and dry in a cool, well-ventilated place.

Requirements for storage rooms and vessels

Containers which are opened must be carefully closed and kept upright to prevent leakage. Always keep in containers of same material as the original one.

7.3 Specific end use(s):

No data available.

Section 8. Exposure controls/Personal protection

8.1 Control parameters:

Occupational exposure limit values

No	Substance name CAS no. EC no.	
1	Dust	
2	List of approved workplace exposure limits (WELs) / EH40	
	total inhalable dust	
	TWA	10 mg/m ³
3	List of approved workplace exposure limits (WELs) / EH40	
	respirable dust	
	TWA	4mg/m ³

8.2 Exposure controls:

Appropriate engineering controls:

Ensure adequate ventilation, local exhaust at the work station if necessary.

Personal protective equipment:

Respiratory protection:

If workplace exposure limits are exceeded, a respiration protection approved for this particular job must be worn. In case of dust formation, take appropriate measures for breathing protection in the event workplace threshold values are not specified.

Respirator P3.

Eye / face protection:

Tightly fitting safety glasses (EN 166).

Hand protection:

In case of intensive contact, wear protective gloves (EN 374). Before use, the protective gloves should be tested in any case for its specific work-station suitability (i.e. mechanical resistance, product compatibility and antistatic properties). Adhere to the manufacturer's instructions and information relating to the use, storage, care and replacement of protective gloves. Protective gloves shall be replaced immediately when physically damaged or worn. Design operations thus to avoid permanent use of protective gloves.

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Appropriate Material	nitrile
Material thickness	>0.3mm
Breakthrough time	>480min.
Appropriate material	rubber
Material thickness	>0.6mm
Breakthrough time	>240min.

Other:

Chemical-resistant work clothes.

Environmental exposure controls:

No data available.

Section 9. Physical and chemical properties

9.1 Information on basic physical and chemical properties:

Form/Colour	
solid	
Almost white, crystalline power or small flakes.	
Odour	
Characteristic	
Odour threshold	
No data available	
PH value	
Value	Appr. 5.6-6.6
Concentration	10% aqueous solution
Source	supplier
Boiling point/boiling range	
No data available	
Melting point / melting range	
Value	Appr. 280°C
Source	supplier
Remarks	Decomposition
Decomposition point / decomposition range	
Value	Appr. 215°C
Source	supplier
Flash point	
No data available	
Auto-ignition temperature	
No data available	
Oxidising properties	
No data available	
Explosive properties	

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No data available	
Flammability (solid, gas)	
No data available	
Lower flammability or explosive limits	
No data available	
Upper flammability or explosive limits	
No data available	
Vapour pressure	
No data available	
Vapour density	
No data available	
Evaporation rate	
No data available	
Relative density	
Density	
No data available	
Solubility in water	
Value	Appr. 29-34g/l
Reference temperature	22-25°C
source	Supplier
Solubility(ies)	
No data available	
Soluble in	
Acids; Bases; Ethanol	
Partition coefficient: n-octanol/water	
No data available	
Viscosity	
No data available	
Solvent content	
value	≤ 0.5%

9.2 Other information:

No data available.

Section 10. Stability and reactivity

10.1 Reactivity:

No data available.

10.2 Chemical stability:

Stable under recommended storage and handling conditions (See section 7).

10.3 Possibility of hazardous reactions:

Dangerous reactions are not to be expected when handling product according to its

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intended use.

10.4 Conditions to avoid:

Air humidity.

10.5 Incompatible materials:

None known.

10.6 Hazardous decomposition products:

No hazardous decomposition products known.

Section 11. Toxicological information

11.1 Information on toxicological effects

Acute oral toxicity
No data available
Acute dermal toxicity
No data available
Acute inhalational toxicity
No data available
Skin corrosion/irritation
No data available
Serious eye damage/irritation
No data available
Respiratory or skin sensitisation
No data available
Germ cell mutagenicity
No data available
Reproduction toxicity
No data available
Carcinogenicity
No data available
STOT-single exposure
No data available
STOT-repeated exposure
No data available
Aspiration hazard
No data available

Section 12. Ecological information

12.1 Toxicity:

Toxicity to fish (acute)
No data available

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Toxicity to fish (chronic)
No data available
Toxicity to Daphnia (acute)
No data available
Toxicity to Daphnia(chronic)
No data available
Toxicity to algae (acute)
No data available
Toxicity to algae(chronic)
No data available
Bacteria toxicity
No data available

12.2 Persistence and degradability:

No data available.

12.3 Bioaccumulative potential:

No data available.

12.4 Mobility in soil:

No data available.

12.5 Results of PBT & vPvB assessment:

No data available.

12.6 Other adverse effects:

No data available.

Section 13. Disposal considerations

13.1 Waste treatment methods:

Product:

Allocation of a waste code number, according to the European Waste Catalogue, should be carried out in agreement with the regional waste disposal company.

Packaging:

Residuals must be removed from packaging and when emptied completely disposed of in accordance with the regulations for waste removal. Incompletely emptied packaging must be disposed of in the form of disposal specified by the regional disposer.

Section 14. Transport information

14.1 Transport ADR/RID/ADN:

The product is not subject to ADR/RID/ADN regulations.

14.2 Transport IMDG:

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The product is not subject to IMDG regulations.

14.3 Transport ICAO-TI/IATA:

The product is not subject to ICAO-TI/IATA regulations.

14.4 Other information:

No data available.

14.5 Environmental hazards:

Information on environmental hazards, if relevant, please see 14.1-14.3.

14.6 Special precautions for user:

No data available.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code:

Not relevant.

Section 15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture EU regulations:

Regulation (EC) No 1907/2006 (REACH) Annex XIV (List of substances subject to authorisation)

In accordance with the Reach regulation (EC) 1907/2006, the product does not contain any substances that are considered as subject to listing in annex XIV, inventory of substances requiring authorization.

REACH candidate list of substances of very high concern(SVHC) for authorization

In accordance with article 57 and article 59 of the Reach regulation (EC) 1907/2006, this substance is not considered as subject to listing in annex XIV, inventory of substances requiring authorisation ("Authorization list").

Regulation (EC) No 1907/2006 (REACH) Annex XVII: RESTRICTIONS ON THE MANUFACTURE, PLACING ON THE MARKET AND USE OF CERTAIN DANGEROUS SUBSTANCES, PREPARATIONS AND ARTICLES

The substance is not subject to the provisions of annex XVII (restriction entries) of the Reach regulation (EC) 1907/2006.

Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances

This substance is not subject to Part 1 or 2 of Annex I

Other regulations

Adhere to the national sanitary and occupational safety regulations when using this product.

15.2 Chemical safety assessment:

For this substance a chemical safety assessment has not been carried out.

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Section 16. Other information

16.1 Further information:

This information is based on our current knowledge however it does not represent a guarantee of product properties nor does it create any legal obligation.

16.2 Sources of key data used to compile the data sheet:

Regulation (EC) No 1907/2006 (REACH), 1272/2008 (CLP) as amended in each case.

EC Directives 2000/39/EC, 2006/15/EC, 2009/161/EU.

National Threshold Limit Values of the corresponding countries as amended in each case.

Transport regulations according to ADR, RID, IMDG, IATA as amended in each case.

The data sources used to determine physical, toxic and ecotoxic data, are indicated directly in the corresponding chapter.

This information is based on our present knowledge and experience.

The safety data sheet describes products with a view to safety requirements.

It does not however, constitute a guarantee for any specific product properties and shall not establish a legally valid contractual relationship.

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